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Nondestructively identifying the mechanical behavior of soft tissues using surface deformation with an explicit inverse approach

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ABSTRACT

Identifying the spatial variation of stiffness properties in soft tissues nondestructively, with limited surface measurements, poses significant challenges. In this paper, we present a novel explicit inverse approach designed to characterize the nonhomogeneous elastic property distribution of soft tissues using only surface displacement datasets. In contrast to the prevalent implicit inverse approach, which focuses on optimizing the elastic properties of individual pixels, our proposed method optimizes the geometric parameters of deformable and movable components, as well as shear moduli of each component. As a result, the proposed approach requires far fewer optimization variables, streamlining the process. Numerical tests conducted in this study demonstrate the superiority of the explicit inverse method over the implicit inverse method, providing much-improved reconstructed results. In particular for a ring structure, while the average relative error of reconstruction using the implicit inverse method can exceed 40 %, the explicit inverse method achieves a remarkable average relative error of only 5 %. Given that surface displacements are easily measurable, the integration of our proposed explicit inverse method with low-cost imaging techniques shows great potential in accurately mapping the elastic property distribution of biological tissues.

Introduction

Biological tissues often display significant heterogeneity. This diversity is evident in various tissues such as skin [1–3], blood vessels [4,5], and corneas [6,7], among others, which exhibit multi-layered structures. These layers offer a variety of functionalities, thus contributing to the complexity of biological systems. Interestingly, the emergence of cancerous tumors introduces an additional level of heterogeneity to these tissues. These tumors typically exhibit greater stiffness compared to their surrounding healthy tissues due to pathological alterations [8]. This stark contrast in mechanical properties can serve as a potential indicator of malignancy. It's also worth noting that the local mechanical properties of soft tissues are not static; they fluctuate in response to various factors such as age

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